

# Assignment 1: Binary Classification \*\*Course:\*\* Data Analysis | \*\*Programme:\*\* M.Tech, Semester I | \*\*Time Allowed:\*\* 1 week \*\*Instructions:\*\* Show all steps for numerical problems. Round to 3 decimal places unless told otherwise. For Q6, submit your code (.py or .ipynb) along with a short report of the outputs. ---

**## Q1. Confusion Matrix and Metrics** A classifier is tested on 200 patients to detect a disease (Positive = diseased). It produces: TP = 40, FP = 10, FN = 20, TN = 130. (a) Draw the confusion matrix. (b) Compute accuracy, precision, recall (sensitivity), specificity, F1-score and false positive rate. (c) Compute balanced accuracy. (d) A "lazy" classifier predicts \*Negative\* for every patient. Compute its accuracy and recall. What does this tell you about using accuracy on imbalanced data? ---

**## Q2. Thresholds, ROC Curve and AUC** A model gives the following predicted probabilities of the positive class for 10 test samples: | Sample | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | |---|---|---|---|---|---|---|---|---|---| | Score | 0.92 | 0.80 | 0.75 | 0.65 | 0.55 | 0.45 | 0.40 | 0.30 | 0.25 | 0.10 | | True label | 1 | 1 | 0 | 1 | 0 | 0 | 1 | 0 | 1 | 0 | | (a) Using threshold 0.5 (predict 1 if score  $\geq 0.5$ ), build the confusion matrix and compute accuracy, precision and recall. (b) Repeat (a) for thresholds 0.7 and 0.35. Tabulate precision, recall and FPR for all three thresholds. (c) Compute (FPR, TPR) for every distinct score used as a threshold and plot the ROC curve by hand or in Python. (d) Compute the AUC using the trapezoidal rule. (e) Interpret the AUC value. Is this a good classifier? ---

**## Q3. Logistic Regression by Hand** Consider a dataset with one feature:  $(x, y) = (1, 0), (2, 0), (3, 1), (4, 1)$ . The model is  $P(y=1 | x) = \text{sigmoid}(w \cdot x + b)$ , where  $\text{sigmoid}(z) = 1 / (1 + e^{-(z)})$ . (a) With  $w = 0, b = 0$ , compute the predicted probability for each point and the binary cross-entropy loss (use natural log). (b) Compute the gradients  $dL/dw$  and  $dL/db$  for the mean loss. (c) Perform one gradient descent update with learning rate 0.1. Report new  $w$  and  $b$ . (d) Recompute the loss after the update. Has it decreased? ---

**## Q4. Decision Boundary of Logistic Regression** A trained model has  $w = 2$  and  $b = -5$ . (a) Find the value of  $x$  at which  $P(y=1 | x) = 0.5$ . (b) Compute  $P(y=1 | x)$  for  $x = 2$  and  $x = 3$ . (c) If the threshold is raised from 0.5 to 0.8, will the decision boundary move left or right? Justify. ---

**## Q5. k-Nearest Neighbours** Training data (2 features): | Point | A | B | C | D | E | F | G | H | |---|---|---|---|---|---|---|---|---| |  $(x_1, x_2)$  | (1,1) | (2,1) | (2,3) | (5,4) | (6,5) | (5,7) | (3,0) | (8,8) | | Class | 0 | 0 | 0 | 1 | 1 | 1 | 1 | 0 | 1 | Classify the query point (4, 4) using Euclidean distance for  $k = 1, 3, 5$  and 8. Comment on the result for  $k = 8$ . ---

**## Q6. Programming Task: Breast Cancer Classification** Use ``sklearn.datasets.load_breast_cancer``. (a) Load the data, report its shape and the class distribution. (b) Split 80/20 into train and test (``random_state=42``, ``stratify=y``). Standardise features using statistics from the training set only. Why must the scaler not be fit on the test set? (c) Train a ``LogisticRegression`` model. Report the confusion matrix, accuracy, precision, recall, F1 and ROC-AUC on the test set. (d) Plot the ROC curve. (e) Train a ``KNeighborsClassifier`` ( $k = 5$ ) on the same split and compare it with logistic regression in a small table. Which would you choose and why? ---

**## Q7. Short Conceptual Questions** (a) In cancer screening, would you prefer high precision or high recall? Explain. (b) Name two techniques to handle class imbalance in binary classification. (c) Why is the F1-score a harmonic mean rather than an arithmetic mean? (d) What is overfitting? Mention one way to detect it and one way to reduce it for a classifier. ---